# Sleep Study on the Job

**An investigation into sleep qualities and occupation.**

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In this report, we aim to derive potential connections between reported sleep quality, and occupation. We will be working with a dataset containing a mixture of occupations, and utilizing sleep quality, sleep duration, and stress levels, to measure possible correlations.

## Importance and Context

We plan to study an answer to the question: "Is there a relationship between sleep quality and different occupations?". This question is interesting to us, as we believe it is a common belief that some occupations demand more hours, or are more stressful than others may be. We wonder if this thought is based in fact, or if people, when studied, claim and behave differently in their twilight hours. There are also related questions we could seek to answer, such as "Does occupation have a notable relation to stress level?" and "Does Occupation have correlation to physical activity level?". The question of whether occupation affects sleep has been studied before, with a mixture of results depending on the study, and individuals polled. Some look at the prevalence of short sleep in certain career fields (Oxford Academic). While others look at associations between working hours and sleep (Taylor & Francis). And some study occupational stress and sleep (NIH). We seek to evaluate a person's claim of sleep quality first and foremost alongside their sleep duration to get a strong measurement of their occupations' possible connection.

To answer this question, we have a data set containing occupations, sleep duration, quality, stress level, physical activity level, and even steps. We can derive information from this, by comparing each occupation and finding unique takeaways from this data set involving sleep duration/quality and career field specifically. We can use linear regression models to find correlations between sleep quality/duration and occupation in our studies, looking in particular for p-values of notable significance. Scatter plots could be useful as well to show trends and potential clustering towards, or away from high sleep quality/duration when considering different occupations. We are excited to immerse ourselves further into this data, and derive potentially notable insights.

## Data and Methodology

In our research, we utilized a data set containing sleep and health/lifestyle data with 374 individual unique Person IDs. We have narrowed this information down further to exclude occupations that we deem to contain too little information to include in our model (There were too few entrants to that occupation). We set the cutoff for this at 20, meaning, we removed occupations that made up around 5% or less of the population. This helped to ensure more accuracy in our models, and brought us down to 7 Occupations out of the original 10. This still left us with a total of 363 rows of data, which we deemed sufficient for this study.Each remaining row contains multiple observations, collected at once from each individual. We specifically would like to focus on the Y-variable of sleep quality, as we noticed unexpected responses when compared to sleep duration6 that created notable diversity between our occupations.Sleep quality is a 1-10 **Y** measurement that we will be treating as a continuous variable in this case. We will also be comparing captured sleep quality and comparing it to sleep duration, stress level, and even physical activity.

Here, we show a model giving a visual of our coefficients for quality of sleep based on occupation.

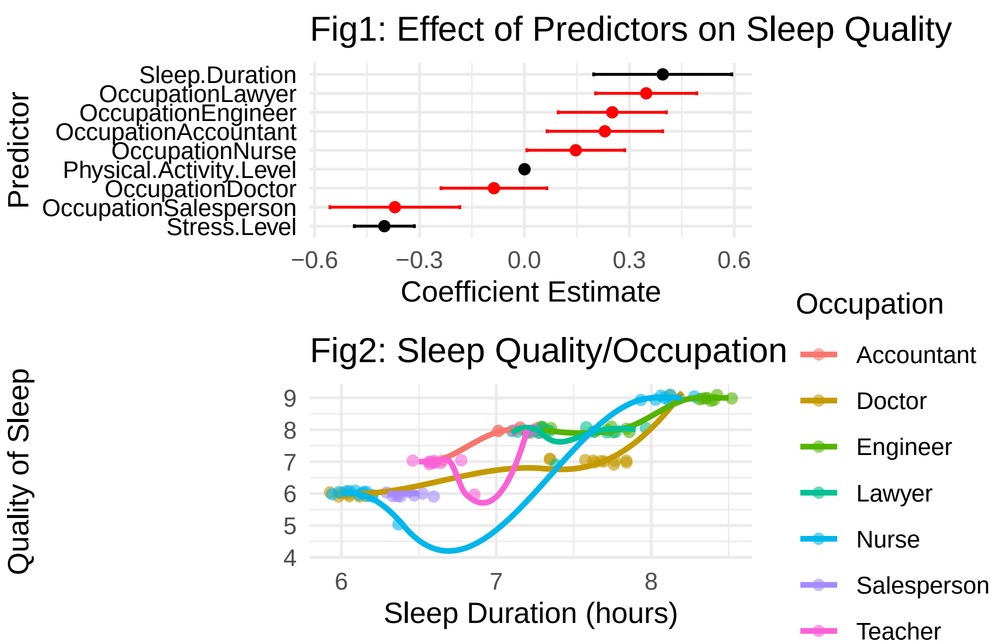
```
Call:
lm(formula = Quality.of.Sleep ~ Occupation, data = exploration_set)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.09524 -0.44444  0.07143  0.19048  2.19048 

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)    
(Intercept)            7.32431    0.08694  84.241  < 2e-16 ***
OccupationAccountant   0.58478    0.23441   2.495  0.01426 *  
OccupationDoctor      -0.51479    0.17995  -2.861  0.00516 ** 
OccupationEngineer     1.12013    0.19110   5.862 6.04e-08 ***
OccupationLawyer       0.60426    0.21164   2.855  0.00524 ** 
OccupationNurse       -0.22908    0.17995  -1.273  0.20599    
OccupationSalesperson -1.32431    0.25588  -5.175 1.19e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8543 on 99 degrees of freedom
Multiple R-squared:  0.4296,    Adjusted R-squared:  0.395 
F-statistic: 12.43 on 6 and 99 DF,  p-value: 2.164e-10
```

From this summary, our our exploratory sleep quality model, we made some important tweaks. As we do not have a baseline occupation, so instead, we took a grand mean to use as our baseline, incorporating all occupations. This number comes to an average of 7.324 quality of sleep. Based off of this data alone, we see a significant p-value for Accountant, Doctor, and Lawyer, very strong significance in both Engineer, and Salesperson, and a value without statistical significance in Nurses. With their coefficients in mind, we can see that some Occupations have a negative correlation, and some positive, with varying degrees. Our R squared gives us a 43% variance in sleep quality and an adjusted value of 39.5% explained by occupation by this slice of data and our number of predictors. With our F-statistic and p-value, we can convincingly imagine that in this data slice, Occupation is a significant predictor of quality of sleep.

```
Call:
lm(formula = Quality.of.Sleep ~ Sleep.Duration + Occupation +
    Stress.Level + Physical.Activity.Level, data = exploration_set)

Residuals:
     Min       1Q   Median       3Q      Max
-1.51769 -0.12771 -0.01268  0.15407  0.67492

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)              6.6297570  0.8771469   7.558 2.42e-11 ***
Sleep.Duration           0.3954567  0.0994568   3.976 0.000136 ***
OccupationAccountant     0.2299514  0.0832190   2.763 0.006861 **
OccupationDoctor        -0.0871412  0.0762434  -1.143 0.255908
OccupationEngineer       0.2511225  0.0778961   3.224 0.001729 **
OccupationLawyer         0.3481317  0.0728712   4.777 6.38e-06 ***
OccupationNurse          0.1467481  0.0704177   2.084 0.039819 *
OccupationSalesperson   -0.3702611  0.0935159  -3.959 0.000144 ***
Stress.Level            -0.4003904  0.0431797  -9.273 5.44e-15 ***
Physical.Activity.Level  0.0003713  0.0017578   0.211 0.833160
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2792 on 96 degrees of freedom
Multiple R-squared:  0.9409,    Adjusted R-squared:  0.9354
F-statistic: 169.8 on 9 and 96 DF,  p-value: < 2.2e-16
```

With the addition of Stress Level, Physical Activity Level, and sleep duration, our model now explains 94.09% of sleep quality variance. We chose to omit age and gender here, and even the existence of sleep disorders.As one may expect, this slice shows that sleep duration and stress level contribute heavily to sleep quality, and possibly surprising to some, physical activity does not appear significant here.

Another is that there seems to be a shift in the effect if occupation on sleep quality, with Doctor no longer appearing significant, and appearing to be more due to stress and/or less sleep duration. These could obviously be affected by occupation, but we are specifically measuring the effect of sleep quality here.




Here is a visual(Figure 1) of our X-variables effects (coefficient) on our outcome variable. We see that sleep duration and stress level are most significant, but when directly evaluating for occupation (in red), Salesperson has a very significant negative correlation, and Lawyer as a very significant positive correlation.

We can see in figure 2, that there is in fact, a positive trend with sleep duration and sleep quality as expected, with some occupations clearly falling behind. We use jittered points here to give a stronger representation of our data visually, and give a properly continuous look. ## Results In this confirmation set, we will first test how well our exploration set can be used to predict our confirmation set.

| | Actual | | | | | | |
|---|---|---|---|---|---|---|---|
| Predicted | Accountant | Doctor | Engineer | Lawyer | Nurse | Salesperson | Teacher |
| Accountant | 20 | 0 | 2 | 9 | 0 | 0 | 4 |
| Doctor | 0 | 48 | 2 | 4 | 1 | 0 | 3 |
| Engineer | 0 | 0 | 24 | 1 | 17 | 0 | 0 |

```
  Lawyer                 4      0        10      17      8             0       0
  Nurse                  2      2         4       2     22             0       0
  Salesperson            0      0         1       0      4            23       3
  Teacher                0      0         2       0      0             0      18
```

When running a prediction on this fitted model, we can see that this model correctly predicts Occupation our key **X** variable (Occupation) at a rate above average in some occupations, but shows extreme accuracy in others such as "Teacher", "Doctor", and "Salesperson". This tells us that this data set may hold some strong trends within professions, and gives us confidence that our data does not suffer from high variance.

We used a similar format below to predict quality of sleep, our **Y**-variable.

```
         Actual
Predicted 1_QoS 2_QoS 3_QoS 4_QoS 5_QoS 6_QoS 7_QoS 8_QoS 9_QoS 10_QoS
   1_QoS      0     0     0     0     0     0     0     0     0      0
   2_QoS      0     0     0     0     0     0     0     0     0      0
   3_QoS      0     0     0     0     0     0     0     0     0      0
   4_QoS      0     0     0     0     0     0     0     0     0      0
   5_QoS      0     0     0     0     4     3     0     0     0      0
   6_QoS      0     0     0     0     0    67     0     0     0      0
   7_QoS      0     0     0     0     1     2    45     0     0      0
   8_QoS      0     0     0     0     0     0     7    72     0      0
   9_QoS      0     0     0     0     0     0     2     3    50      0
   10_QoS     0     0     0     0     0     0     0     0     0      0
```

We can see similarly here, that Quality of Sleep (QoS) also has strong predictability. This is a very good sign that our data is significant.

Next, we

```
             Df Sum Sq Mean Sq F value Pr(>F)
Occupation    6  194.8   32.46   41.65 <2e-16 ***
Residuals   356  277.5    0.78
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = Quality.of.Sleep ~ Occupation, data = dat)
```

```
$Occupation
                                diff         lwr          upr     p adj
Doctor-Accountant      -1.244004568 -1.77482611 -0.713183023 0.0000000
Engineer-Accountant     0.520806521 -0.02143849  1.063051530 0.0690249
Lawyer-Accountant       0.001725129 -0.57365724  0.577107501 1.0000000
Nurse-Accountant       -0.522028878 -1.05035338  0.006295622 0.0552002
Salesperson-Accountant -1.891891892 -2.52388944 -1.259894342 0.0000000
Teacher-Accountant     -0.916891892 -1.51403882 -0.319744960 0.0001469
Engineer-Doctor         1.764811089  1.31168439  2.217937786 0.0000000
Lawyer-Doctor           1.245729697  0.75342996  1.738029431 0.0000000
Nurse-Doctor            0.721975690  0.28560322  1.158348158 0.0000291
Salesperson-Doctor     -0.647887324 -1.20530553 -0.090469118 0.0112477
Teacher-Doctor          0.327112676 -0.19045723  0.844682586 0.4989307
Lawyer-Engineer        -0.519081391 -1.02367743 -0.014485350 0.0391550
Nurse-Engineer         -1.042835399 -1.49303432 -0.592636478 0.0000000
Salesperson-Engineer   -2.412698413 -2.98100572 -1.844391107 0.0000000
Teacher-Engineer       -1.437698413 -1.96697787 -0.908418951 0.0000000
Nurse-Lawyer           -0.523754008 -1.01336027 -0.034147745 0.0272005
Salesperson-Lawyer     -1.893617021 -2.49362404 -1.293610007 0.0000000
Teacher-Lawyer         -0.918617021 -1.48179725 -0.355436790 0.0000403
Salesperson-Nurse      -1.369863014 -1.92490384 -0.814822185 0.0000000
Teacher-Nurse          -0.394863014 -0.90987163  0.120145604 0.2598364
Teacher-Salesperson     0.975000000  0.35409099  1.595909012 0.0000925
```

```
[1] NA
```

Evaluate accuracy or confusion matrix table(Predicted = confirmation_set$Predicted, Actual = confirmation_set$Occupation)

## Discussion

## Final Report

Your final report should document your analysis, communicating your findings in a way that is technically precise, clear, and persuasive.

Page limits:

- Main report: 4 pages
- Appendix: 1 additional page

You must meet these page limits using standard `pdf_document` output in a `.Rmd` file, or in a `documentclass: scrartcl` in a `.qmd` file. (This assignment prompt has been generated as a `scrartcl`).

The one-page appendix is intended for extra information that will help your instructor assess your model building process. Please include the following elements:

1. **A Link to your Data Source.** If you used specialized code to access your data, please include that here. Please make sure your instructor has the ability to access the data.
2. **A List of Model Specifications you Tried.** We are interested in seeing how you arrived at your final model. In just a sentence, please provide a reason or something that you learned from each specification.
3. **A Residuals-vs-Fitted-values Plot.** Please generate this plot for your final model that includes variable transformations. Your instructor will use this plot to assess how well you have captured the shape of the relationship between your X and your Y variable. For example, if there is a clear parabolic pattern in your residuals-vs-fitted plot, that is a signal that you should have included a square term.

## Evaluation Criteria

We present the following criteria to guide you to a professional-quality report. Moreover, these criteria are also the ones we will use to grade your report. The descriptions below are copied directly from our grading rubric.

### 1. Intro

An introduction that is scored in the top level has very successfully made the case for the study. It will have explained why the topic is interesting, and provided compelling reasons to care about not just the general area, but rather about every concept in the research question and the statistical results to be generated. The introduction will be engaging from the very first sentence, and create a logical story that leads the reader step-by-step to the research question. After reading the introduction, no part of the research question will appear arbitrary or unexpected, instead flowing naturally from the background provided.

### 2. Description of the Data Source

A report that is scored in the top level will describe the provenance of the data; the audience will know the source of the data, the method used to collect the data, the units of observation of the data, and important features of the data that are useful for judging the analysis.

### 3. Data Wrangling

A report that is scored in the top level on data wrangling will have succeeded to produce a modern, legible data pipeline from raw data to data for analysis. Because there are many pieces of data that are being marshaled, the reports that score in the top level will have refactored different steps in the data handling into separate files, functions, or other units. It should be clear what, and how, any additional features are derived from this data. The analysis avoid defining additional data frames when a single source of truth data would suffice.

### 4. Operationalization

A report that is scored in the top level on operationalization will have precisely articulated and justified the decisions leading to the variables used in the analysis. The reader will be left with a clear understanding of the concepts in the research question, and how they relate to the operational definitions, including any major gaps that may impact the interpretation of results. You should also list how many observations you remove and for what reasons. When there is more than one reasonable way to operationalize a concept, the report will explain the alternatives and provide reasons for the one that was selected. Note that there is often more than one way to operationalize a concept; we are less interested in whether you make the best possible choice, and more in how well you defend your decisions.

### 5. One or More Data Visualization(s)

This is a task of description, and one of the most effective ways to describe relationships between variables is through visualization.

You are required to include at least one plot in your main report that highlights the relationship between your $Y$ and your $X$ variables In your text, you should provide your interpretation of what this plot means, and link to the plot. For example, we might write that Figure 1 displays 300 observations from a synthetic model of Earnings and Age, showing an increasing trend though the relationship is less positive for higher ages.

Include a visual representation of your final model predictions on the plot. A report that is scored in the top level will have plots that effectively transmit information, engage the reader's interest, maximize usability, and follow best practices of data visualization. Titles and labels will be informative and written in plain English, avoiding variable names or other artifacts of R code. Plots will have a good ratio of information to space or information to ink; a large or complicated plot will not be used when simple plot or table would show the same information more directly. Axis limits will be chosen to minimize visual distortion and avoid misleading the viewer. Plots will be free of visual artifacts created by binning. Colors and line types will be chosen to reinforce the meanings of variable levels and with thought given to accessibility for the visually-impaired. Your report will be free of "output dumps" - code output that has not been formatted for human-readability. Every single plot and table in the report will be discussed in the narrative of the report.

### 6. Model Specification

A report that is scored in the top level will have chosen a set of regression models that strongly support the goal of the study. Variables transformations will be chosen to inform the reader of the shape of the joint distribution, and will be human-understandable. A reason will be provided for the chosen variable transformations. Ordinal variables will not be treated as metric. All model specifications will be displayed in a regression table, using a package like `stargazer` to format your output. Displayed standard errors will be correctly chosen.

#"'{r report models, message = FALSE, warning = FALSE, results = 'asis'} ##| results: asis ##| warning: FALSE ##| message: FALSE ##| tbl-cap: Regression Results. ##| #| tbl-pos: t #stargazer( # model_one, # model_two, # model_three, # header = FALSE, # se = list( ## sqrt(diag(vcovHC(model_one))), # sqrt(diag(vcovHC(model_two))), # sqrt(diag(vcovHC(model_three))) # ), # type = ifelse(knitr::is_latex_output(), "latex", "html"), # covariate.labels = c("Age", "

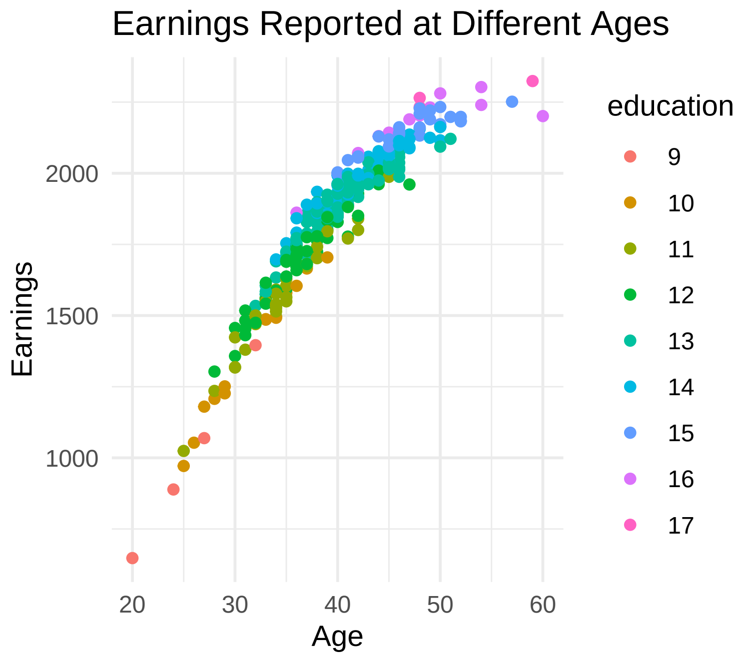



Figure 1: Synthetic data showing a hypothetical relationship between Earnings, Age, and Education. An increasing trend can be seen with decreasing slope for higher ages.

$textAge^2$", "Education", "Intercept"), # digits = 2, # omit.stat = c('ser', 'f'), # star.cutoffs = c(0.05, 0.01, 0.001) #)

"'

### 7. Model Assumptions

A report that scores in the top-level has provided an thorough and precise assessment of the assumptions supporting the regression. The list of assumptions is appropriate given the sample size and modeling goals. Each assumption is evaluated fairly, and discussed defensively - without overstating how credible the assumption is or rendering a final up-or-down judgement on the assumption. The report will not miss any important violations of any assumption. Where possible, the report discusses the consequences for the analysis of a violated assumption.

### 8. Model Results and Interpretation

A report that scores in the top level will correctly interpret statistical significance, clearly interpret practical significance, and comment on the broader implications of the results. It may want to include statistical tests besides the standard t-tests for regression coefficients. When discussing practical significance, comment on both the direction and magnitude of your coefficients, placing them in context so the reader can understand if they are important. To help the reader understand your fitted model, you may want to describe hypothetical datapoints (e.g. a hypothetical person with 1 cat is predicted to spend $2400 on pet care. That rises to $3200 for a hypothetical person with 2 cats…).

### 9. Overall Effect

A report that scores in the top level will have met expectations for professionalism in data-based writing, reasoning and argument for this point in the course. It can be presented, as is, to another student in the course, and that student could read, interpret and take away the aims, intents, and conclusions of the report.

## References